

The examination is being carried out on the **following application documents**

Description, Pages

1-175 as originally filed

Drawings, Sheets

1/8-8/8 as originally filed

Cited documents

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

- D1 WO 2008/039218 A2 (PHARMACYCLICS INC [US]; HONIGBERG LEE [US]; VERNER ERIK [US]; PAN ZHEN) 3 April 2008 (2008-04-03)
- D2 WO 2007/136790 A2 (MANNKIND CORP [US]; FLYNN GARY A [US]; LEE SANDRA AEYOUNG [US]; FARIS) 29 November 2007 (2007-11-29)

The prior art

D1 (WO2008039218) claims, in particular claim 8; §0041; compounds of §0205; §00412, table 3

discloses compounds of present formula I, preferably compounds of claim 8 of D1, as inhibitors of Btk for use in the treatment of cancer, preferably a B-cell proliferative disorder such as lymphoma (§0034), e.g. Burkitt's lymphoma (= a solid tumor). It is explicitly disclosed/suggested to treat pancreatic cancer and other solid or hematological tumors (§0041). Compound 4, falling under formula I was shown to inhibit also further tyrosine kinases, e.g. Lck, Lyn, Itk (§00412, table 3). Anticipates the subject-matter of claims 1-6, 8-13, 15-18, 22 and possibly claims 23-26.

D2 (WO2007136790) claims; [55], [110]

discloses irreversible TKI inhibitors, preferably of inhibitors of BTK and ITK. The irreversible inhibition is achieved via attachment per Michael reaction on cysteine at the active site, see e.g. [0142], [0146]. The compounds are disclosed to treat inter alia

solid tumors [07], such as **lymphomas** (i.e. solid tumors), such as **Non-Hodgkin lymphoma** [110]. The compounds also seem to inhibit EGFR, Her 2, Her 4, and Jak kinase (e.g., Jak3), see [55]. Compounds were shown to inhibit ITK, Btk and LCK (table 2, p. 71). Anticipates the subject-matter of claims 1, 6, 9, 15-18, 22.

Art. 82 EPC - Lack of unity of invention

According to Article 82 EPC, an application shall relate to one invention only or a group of inventions linked by one or more of the same or corresponding technical features. The present application is considered to relate to 3 inventions for the following reasons.

I: Claims 1-14 and partial claims 15-26 directed to a method of treatment wherein a composition is used comprising a compound that forms a covalent bond with a cysteine sidechain of a Btk or Btk homolog or BTK cysteine homolog.

II: Partial claims 15-26 directed to a method of treatment wherein a composition is used comprising a compound that forms a covalent bond with a cysteine sidechain of HER4.

III: Partial claims 15-26 directed to a method of treatment wherein a composition is used comprising a compound that forms a covalent bond with a cysteine sidechain of an ACK other than that of inventions I and II.

Said inventions are not so linked as to form a single general inventive concept (Rule 44 EPC) for the following reasons.

The problems underlying the present inventions I to III is the treatment of one or more solid tumors or developing solid tumor in a patient.

The solutions proposed are the following:

ad I) the therapeutic use of a composition comprising a compound that forms a covalent bond with a cysteine sidechain of a Btk or Btk homolog or BTK cysteine homolog in said diseases.

ad II) the therapeutic use of a composition comprising a compound that forms a covalent bond with a cysteine sidechain of HER4 in said diseases.

ad III) the therapeutic use of a composition comprising a compound that forms a covalent bond with a cysteine sidechain of an ACK other than that of inventions I and II.

The common concept linking all inventions I-III is that a composition comprising a compound that forms a covalent bond with a cysteine sidechain of an ACK (such as e.g. Btk or Her4 or any further ACK thereby inhibiting irreversibly said enzyme) is used to treat a solid tumor or a developing solid tumor.

This concept is not novel for the following reasons:

D1 (WO2008039218) claims, in particular claim 18-21; §0041; compounds of §0205; §00412, table 3

discloses compounds of present formula I, compounds of claim 8 of D1 or table on p. 74/75 of D1: compounds 4-15, as irreversible inhibitors of Btk for use in the treatment of cancer, preferably a B-cell proliferative disorder such as lymphoma (§0034, a solid tumor, see e.g. claim 6 of present application), e.g. Burkitt's **lymphoma** (= a solid tumor). It is furthermore explicitly disclosed that said compounds also treat **pancreatic cancer** and other **solid** or hematological **tumors** (§0041). These irreversible Btk inhibitors preferably bind covalently to a cystein 481 residue of the Btk (Fig. 1, [00139], [00176], [00415], claims 18-20) via a Michael reaction, see [0120], [0121], [00176], [00177], [00411] [00415], [00409]. Further tyrosine kinases carrying cysteine in position 481 such as e.g. ITK (see Fig. 1) are also taught to be inhibited by the compounds of D1 [00139], [00155].

Likewise, D2 (WO2007136790: claims; [55], [110], [0142], [0146], table 2) discloses irreversible TKI inhibitors, preferably of inhibitors of BTK and ITK. The irreversible inhibition is achieved via attachment per Michael reaction on cysteine at the active site, see e.g. [0142], [0146]. The compounds are disclosed to treat inter alia **solid tumors** [07], such as **lymphomas** (i.e. solid tumors), such as **Non-Hodgkin lymphoma** [110]. The compounds also seem to inhibit EGFR1, Her 2, Her 4, and Jak kinase (e.g., Jak3), see [55]. Compounds were shown to inhibit ITK, Btk and LCK (table 2, p. 71).

In light of any of these documents, the common concept identified above is not novel.

Thus, in view of D1 and D2, cited above, the inventions I-III are not so linked as to form a single general inventive concept, i.e. Art. 82 and Rule 44 EPC have not been fulfilled. As a consequence, the application is considered to relate to 3 separate inventions as identified above, of which the first has been examined with the following remarks made here below.

Art. 76 EPC

The applicant is kindly requested to provide a basis for the presently claimed subject-matter (or that of the later amended claim set) in the parent application.

Art. 83 EPC

1.) Claims 1-6-8,15 are not sufficiently disclosed and unclear as they relate to compounds that "form a covalent bond with a cysteine side chain of a Bruton's tyrosine kinase or a Btk cysteine homolog" or "ACK" or "HER4". The skilled person is not aware how to find further compounds with this required functionality. Undue experimentation would be required to screen compounds having the sought activity randomly (Art. 83 EPC) or to test potential Michael reactants if they actually covalently bind to the claimed proteins. Similarly, the feature of claim 3 "[the substituent] is a moiety that binds to the active site of Btk" leaves the skilled person in doubt which moiety falls under this definition (Art. 84 EPC) and which compounds might be used as a starting point for the screening of such moieties (Art. 83 EPC).

2.) Claim 23 encompasses an undetermined number of compounds which are defined only by reference to a desired functional activity, namely the compounds being "**prodrugs**". This functional term does not give a specific technical guidance for the ascertainment of the scope of the claim as to which derivatives are suitable "prodrugs" having the desired function without imposing an unreasonable burden, and thus can be seen as a mere invitation to the skilled person to perform a research program in order to find the suitable variants in contravention of Article 84 EPC (cf. T 279/07, point 2.1 of the decision). Further, there is no general common knowledge of this kind available to the skilled person, and the application does not provide sufficient disclosure (Article 83 EPC).

Art. 84 EPC (Clarity)

The drawing defining formula (VII) is illegible throughout the claims and description. It is accordingly not clear how the substituents Y, Z, R6 to R8 are attached to circle drawn therein. Since claims 24-26 seem to be dependent thereon, even if the claim numbers of claims 24-25 wrongly refer back to claims 21 or 22, **no reasonable opinion may expressed in respect of these claims.**

Art. 54(1) and (2) EPC (Novelty)

The present application does not meet the requirements of Article 52(1) EPC, because the subject-matter of claims 1-6, 8-13, 15-18, 22 (and possibly claims 23-26, see comment above) is not new in the sense of Article 54(1) and (2) EPC in view of D1 and D2, see above.

In respect of independent claim 9 it is noted that D1 does not explicitly disclose the mechanism of action of inhibiting HER4. However, the claim neither requires that the solid tumor is a Her4-dependent solid tumor nor that the activity on solid tumor requires inhibition of Her4. Thus, it appears that the discovery of novel mechanism of action does not translate into a novel therapeutic use, at least not for the whole scope of said claim. Accordingly, D1 is considered to be prejudicial to the novelty of the respective claims.

Art. 56 EPC (Inventive step)

The present application does not meet the requirements of Article 52(1) EPC, because the subject-matter of claims 7 and 14 does not involve an inventive step in the sense of Article 56 EPC.

D1 may be considered to be the closest prior art as it discloses compounds of present claims for use in the treatment of solid tumors, e.g. Burkitt's lymphoma (§0029) or pancreatic cancer (§0041). Claims 7, 14 differ therefrom in that they relate to certain specific solid cancers such as carcinoma or melanoma.

Thus, the problem to be solved in view of D1 is to find an alternative tumor which might be susceptible to the treatment with an irreversible Btk inhibitor.

In this context it is noted that no data or other evidence were provided showing that a compound of formula I or any other Btk inhibitor actually treats the *claimed* solid tumor types. Thus, the applicant extrapolated from the antitumor-activity of Btk inhibitors against a variety of B-cells (Fig. 1 A of present application) to the anti-tumor activity of said compounds to other solid cancer types, in particular those of claims 7 and 14.

By applying the same level of skill, the skilled person would extrapolate from the anti-tumor activity of the claimed compounds as disclosed in D1 to the usefulness of said compounds in the treatment of melanoma or carcinoma, which are both commonly known solid tumors.

The subject-matter of the claims is obvious in view of D1 taken alone or combined with the teaching of D2 that Btk and/or HER4-inhibition treats solid tumors mentioned in D2.

Final remarks

When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

Amendments should be made by filing replacement pages. Unnecessary recasting of the description should be avoided. An amended abstract is not required. The applicant should also take account of the requirements of Rule 50(1) EPC. If handwritten amendments are submitted, they should be clearly legible to the printer.